Exercise 1 — TXT File (Login Logs)

File: logins.txt

Rahul
Sneha
Rahul
Arjun
Sneha
Rahul
Karan

Questions

1. Read the file and print all names.
2. Count the total number of login records.
3. Find how many times each user logged in.
4. Find the user who logged in the most.
5. Print the unique users.

```
file=open("logins.txt","r")
lines=file.readlines()
file.close()
names=[]
for line in lines:
    names.append(line.strip())
print(names)
print(len(names))
count={}
for name in names:
    if name in count:
        count[name]+=1
    else:
        count[name]=1
print(count)
max_user=max(count,key=count.get)
print(max_user)
print(set(names))
```

Exercise 2 — TXT File (Numbers Dataset)

File: numbers.txt

45
67
23
89
12
90
56

34
78
21

Questions

1. Read all numbers from the file.
2. Calculate the sum of all numbers.
3. Find the maximum number.
4. Find the minimum number.
5. Count how many numbers are greater than 50.

```
file=open("numbers.txt","r")
lines=file.readlines()
file.close()
numbers=[]
for line in lines:
    numbers.append(int(line.strip()))
print(numbers)
print(sum(numbers))
print(min(numbers))
count=0
for num in numbers:
    if num>50:
        count+=1
print(count)
```

Exercise 3 — JSON File (Student Dataset)

File: students.json

```
{
"students": [
{"name": "Rahul", "marks": 85, "course": "Python"},
{"name": "Sneha", "marks": 92, "course": "Data Science"},
{"name": "Arjun", "marks": 78, "course": "Python"},
{"name": "Priya", "marks": 88, "course": "AI"},
{"name": "Karan", "marks": 70, "course": "Python"}
]
}
```

Questions

1. Print all student names.
2. Print students enrolled in Python course.
3. Find the student with highest marks.
4. Calculate average marks.
5. Count how many students are enrolled in each course.

```python
import json
from collections import defaultdict
with open("students.json","r") as file:
    data=json.load(file)
students=data["students"]
for s in students:
    print(s["name"])
for s in students:
    if s["course"]=="Python":
        print(s["name"])
top_student=max(students,key=lambda x:x["marks"])
print(top_student["name"],top_student["marks"])
avg_marks=sum(s["marks"] for s in students)/len(students)
print(avg_marks)
course_count=defaultdict(int)
for s in students:
    course_count[s["course"]]+=1
for course,count in course_count.items():
    print(course,":",count)
```

Exercise 4 — JSON File (E-commerce Orders)

File: orders.json


```
{
"orders": [
{"order_id": 1, "customer": "Rahul", "amount": 2500},
{"order_id": 2, "customer": "Sneha", "amount": 1800},
{"order_id": 3, "customer": "Rahul", "amount": 3200},
{"order_id": 4, "customer": "Arjun", "amount": 1500},
{"order_id": 5, "customer": "Sneha", "amount": 2100}
]
}
```


Questions

1. Print all orders.
2. Calculate total revenue.
3. Find total spending per customer.
4. Find the highest spending customer.
5. Count total orders per customer.

```python
import json
from collections import defaultdict
with open("orders.json","r") as file:
    data=json.load(file)
orders=data["orders"]
for order in orders:
    print(order)
total_revenue=sum(order["amount"] for order in orders)
print(total_revenue)
spending=defaultdict(int)
for order in orders:
    spending[order["customer"]]+=order["amount"]
for customer,amount in spending.items():
    print(customer,":",amount)
highest_customer=max(spending,key=spending.get)
print(highest_customer)
order_count=defaultdict(int)
for order in orders:
    order_count[order["customer"]] +=1
```

```python
for order in orders:
    order_count[order["customer"]] +=1
for customer,count in order_count.items():
    print(customer,":",count)
```

Exercise 5 — CSV File (Employee Dataset)

File: employees.csv

name,department,salary
Rahul,IT,70000
Sneha,HR,60000
Arjun,IT,75000
Priya,Finance,80000
Karan,IT,72000

Questions

1. Print all employee names.
2. Find employees working in IT department.
3. Calculate the average salary.
4. Find the highest salary employee.
5. Count how many employees belong to each department.

```python
import csv
from collections import defaultdict
employees=[]
with open("employees.csv","r") as file:
    reader=csv.DictReader(file)
    for row in reader:
        row["salary"]=int(row["salary"])
        employees.append(row)
for emp in employees:
    print(emp["name"])
for emp in employees:
    if emp["department"]=="IT":
        print(emp["name"])
avg_salary=sum(emp["salary"] for emp in employees)/len(employees)
print("\n Average salary:",avg_salary)
highest=max(employees,key=lambda x: x["salary"])
print(highest["name"],"-",highest["salary"])
dept_count=defaultdict(int)
for emp in employees:
```

```python
for emp in employees:
    dept_count[emp["department"]]+=1
for dept,count in dept_count.items():
    print(dept,":",count)
```

Exercise 6 — CSV File (Sales Dataset)

File: sales.csv

product,quantity,price
Laptop,5,75000
Mouse,20,500
Keyboard,15,1500
Laptop,3,75000
Mouse,10,500

Questions

1. Calculate total sales revenue.
2. Find total quantity sold per product.
3. Find the product with highest sales.
4. Calculate total revenue per product.

5. Print products with sales above
50,000.

```python
import csv
from collections import defaultdict
from orders import total_revenue
sales=[]
with open("sales.csv","r") as file:
    reader=csv.DictReader(file)
    for row in reader:
        row["quantity"]=int(row["quantity"])
        row["price"]=int(row["price"])
        sales.append(row)
total_revenue=sum(item["quantity"] * item["price"] for item in sales)
qty_per_product=defaultdict(int)
for item in sales:
    qty_per_product[item["product"]] += item["quantity"]
revenue_per_product=defaultdict(int)
for item in sales:
    revenue_per_product[item["product"]]+=item["quantity"] * item["price"]
highest_product=max(revenue_per_product,key=revenue_per_product.get)
high_sales=[p for p, rev in revenue_per_product.items() if rev>50000]
```

Activate Windows

```python
highest_product=max(revenue_per_product,key=revenue_per_product.get)
high_sales=[p for p, rev in revenue_per_product.items() if rev>50000]
for product in qty_per_product:
    print(f"{product}-> qty:{qty_per_product[product]} Revenue:{revenue_per_product[produ
```

Bonus Challenge

Using the sales.csv file:
Write Python code to produce this output:

Product Sales Summary
Laptop → Qty: 8 Revenue: 600000
Mouse → Qty: 30 Revenue: 15000

Keyboard → Qty: 15 Revenue:
22500

```python
import csv
from collections import defaultdict
qty=defaultdict(int)
revenue=defaultdict(int)
with open("sales.csv","r") as file:
    reader=csv.DictReader(file)
    for row in reader:
        product=row["product"]
        quantity=int(row["quantity"])
        price=int(row["price"])
        qty[product]+=quantity
        revenue[product]+=quantity * price
for product in qty:
    print(f"{product} -> Qty:{qty[product]} Revenue:{revenue[product]}")
```